

Playwright Stabilization Assessment Report

Submitted By

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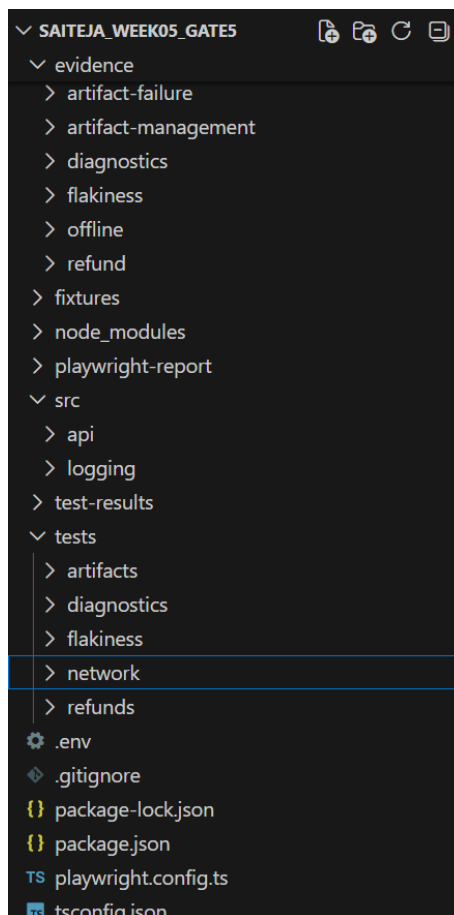
Technology: Playwright with TypeScript

Repository: [ust-sdet-training/SDET at saiteja_week05_gate5](#)

Note:

Meaningful comments have been added throughout the test files to improve readability and make the test flow easier to understand. This report presents each assessment scenario using the **Diagnose → Fix → Evidence → Explain** approach, supported by relevant screenshots.

Code Structure:



1. Objective

The objective of this assessment is to stabilize the Playwright automation suite by identifying failures, diagnosing their root causes, validating fixes, and providing supporting evidence. The assessment focuses on writing reliable tests and demonstrating debugging maturity.

2. Assessment Overview

This assessment covers the following topics:

- Flakiness Analysis
- Diagnostic Logging
- Artifact Management
- Offline & Resilient UI
- Returns & Refund Testing

The overall approach followed throughout the assessment is:

Diagnose → Fix → Evidence → Explain

3. Tools Used

- Playwright
 - TypeScript
 - Node.js
 - Visual Studio Code
 - Git & GitHub
-

4. Flakiness Analysis

Command to run in terminal:

```
npx playwright test makeltFlaky.spec.ts --repeat-each=50
```

Test File: makeltFlaky.spec.ts

Scenario 1 – Race Condition

Issue

The cart total was verified before the asynchronous request completed.

Diagnosis

The assertion executed before the UI finished updating, causing intermittent failures.

Fix

Use Playwright's web-first assertions and wait for the UI state instead of using very short timeouts.

Evidence

- Screenshot of failed test (Before)

```
15     await page.getByRole("button", { name: "Refresh cart total" }).click();
16
17     // Check the cart total immediately, which is the flaky part.
18     await expect(page.getByTestId("debug-cart-total")).toContainText("Rs. 9,197", {
19         timeout: 620
20     });
21 });
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  PLAYWRIGHT
before the async request has settled
4 flaky
[chromium] > tests\flakiness\makeItFlaky.spec.ts:10:7 > Intentionally flaky example > bad example: asserts cart total
before the async request has settled
[chromium] > tests\flakiness\makeItFlaky.spec.ts:10:7 > Intentionally flaky example > bad example: asserts cart total
before the async request has settled
[chromium] > tests\flakiness\makeItFlaky.spec.ts:10:7 > Intentionally flaky example > bad example: asserts cart total
before the async request has settled
[chromium] > tests\flakiness\makeItFlaky.spec.ts:10:7 > Intentionally flaky example > bad example: asserts cart total
before the async request has settled
13 passed (1.6m)
Serving HTML report at http://localhost:9323. Press Ctrl+C to quit.
```

- Screenshot of stable execution (After)

```
16
17     // Check the cart total immediately, which is the flaky part.
18     await expect(page.getByTestId("debug-cart-total")).toContainText("Rs. 9,197");
19 });
20
```

```
C:\Users\299172\Documents\Saiteja_week05_gate5>npx playwright test makeItFlaky.spec.ts
Running 2 tests using 2 workers
2 passed (10.2s)
To open last HTML report run:
npx playwright show-report
```

- HTML Report screenshot (Before)

⚠ Intentionally flaky example › bad example: asserts cart total before the async request has settled	chromium	13.0s
flakiness/makeltFlaky.spec.ts:10	View Trace	
⚠ Intentionally flaky example › bad example: asserts cart total before the async request has settled	chromium	15.7s
flakiness/makeltFlaky.spec.ts:10	View Trace	
⚠ Intentionally flaky example › bad example: asserts cart total before the async request has settled	chromium	10.1s
flakiness/makeltFlaky.spec.ts:10	View Trace	
⚠ Intentionally flaky example › bad example: asserts cart total before the async request has settled	chromium	9.0s
flakiness/makeltFlaky.spec.ts:10	View Trace	
✓ Intentionally flaky example › bad example: asserts cart total before the async request has settled	chromium	5.6s
flakiness/makeltFlaky.spec.ts:10		
✓ Intentionally flaky example › bad example: brittle product locator depends on card position	chromium	5.8s
flakiness/makeltFlaky.spec.ts:23		
✓ Intentionally flaky example › bad example: brittle product locator depends on card position	chromium	6.1s
flakiness/makeltFlaky.spec.ts:23		
✓ Intentionally flaky example › bad example: brittle product locator depends on card position	chromium	7.6s
flakiness/makeltFlaky.spec.ts:23		
✓ Intentionally flaky example › bad example: asserts cart total before the async request has settled	chromium	6.8s
flakiness/makeltFlaky.spec.ts:10		
✓ Intentionally flaky example › bad example: brittle product locator depends on card position	chromium	3.3s
flakiness/makeltFlaky.spec.ts:23		
✓ Intentionally flaky example › bad example: brittle product locator depends on card position	chromium	4.7s

- HTML Report screenshot (After)

flakiness/makeltFlaky.spec.ts		
✓ Intentionally flaky example › bad example: asserts cart total before the async request has settled	chromium	5.2s
flakiness/makeltFlaky.spec.ts:10		
✓ Intentionally flaky example › bad example: brittle product locator depends on card position	chromium	5.0s
flakiness/makeltFlaky.spec.ts:21		

Scenario 2 – Brittle Locator

Issue

The locator depended on the position of a product card.

Diagnosis

Any UI order change caused the test to fail.

Fix

Use stable locators such as `getByRole()` or `getByTestId()`.

Evidence

- Screenshot of failure (Before)

```

24
25 // Navigate to a product using a fragile position-based selector.
26 await page.locator(".product-grid article:nth-child(2) a").click();
27
28 // Assert that the expected product page is visible.
29 await expect(
30   page.getByRole("heading", { name: "Travel Backpack" })
31 ).toBeVisible();

```

- Screenshot after using stable locator (After)

```

24
25 // Navigate to a product using a fragile position-based selector.
26 await page.getByRole("link", { name: "View Travel Backpack" }).click();
27
28 // Assert that the expected product page is visible.
29 await expect(
30   page.getByRole("heading", { name: "Travel Backpack" })
31 ).toBeVisible();
32 });

```

5. Diagnostic Logging

Command to run in terminal:

npx playwright test diagnostics-logging.spec.ts

Test File: diagnostics-logging.spec.ts

Purpose

To improve debugging by capturing structured logs for every important step in the checkout flow.

Implemented

- Correlation ID
- Structured logging
- Response time logging
- Sensitive data redaction

Benefits

- Easy request tracing
- Faster debugging
- Secure logging without exposing sensitive information

Evidence

- Correlation ID and Redacted log screenshot of the terminal

```
...s\diagnostics\diagnostics-logging.spec.ts:12:7 > Diagnostics and Logging > records a correlated checkout diagnostic trail
[INFO] 2026-07-04T11:27:35.007Z - test started
[INFO] 2026-07-04T11:27:35.020Z - checkout journey started {"cartSession":"w5d2-diagnostics-1783164455019"}
...sts\diagnostics\diagnostics-logging.spec.ts:113:7 > Diagnostics and Logging > redacts sensitive fields in structured logs
[INFO] 2026-07-04T11:27:35.098Z - test started
[INFO] 2026-07-04T11:27:35.103Z - payment payload prepared {"cardNumber":"[REDACTED]","headers":{"authorization":"[REDACTED]"},"orderId":"ORD-LOG-1001","token":"[REDACTED]"}
[INFO] 2026-07-04T11:27:35.123Z - test finished {"status":"passed"}
...s\diagnostics\diagnostics-logging.spec.ts:12:7 > Diagnostics and Logging > records a correlated checkout diagnostic trail
[INFO] 2026-07-04T11:27:35.132Z - cart item added {"cartSession":"w5d2-diagnostics-1783164455019","durationMs":112,"httpStatus":201,"productId":101}
[INFO] 2026-07-04T11:27:35.205Z - order placed {"cartSession":"w5d2-diagnostics-1783164455019","durationMs":24,"httpStatus":201,"orderId":5001,"orderNumber":"ORD-1009","total":4598}
[INFO] 2026-07-04T11:27:35.254Z - test finished {"status":"passed"}
2 passed (7.2s)
```

- HTML Report

Project: chromium		4/7/2026, 4:57:30 pm	Total time: 7.2s
▼ diagnostics/diagnostics-logging.spec.ts			
✓	Diagnostics and Logging > records a correlated checkout diagnostic trail	chromium	2.3s
	diagnostics/diagnostics-logging.spec.ts:12		
✓	Diagnostics and Logging > redacts sensitive fields in structured logs	chromium	2.3s
	diagnostics/diagnostics-logging.spec.ts:113		

6. Artifact Management

Command to run in terminal:

npx playwright test artifact-management.spec.ts

npx playwright test artifact-failure.spec.ts

Test Files

- artifact-management.spec.ts
- artifact-failure.spec.ts

Purpose

To capture useful debugging evidence whenever tests pass or fail.

Artifacts Captured

- Playwright HTML Report
- Failure Screenshot
- Cart Evidence

Benefits

Developers can understand failures without reproducing the issue.

Evidence

- HTML Report (Success)

captures cart evidence without attaching it on a passing test

artifacts/artifact-management.spec.ts:13 2.1s

chromium

✓ Run

Test Steps

Filter steps

> ✓ Before Hooks	1.2s
> ✓ Navigate to "/debug-lab" — artifacts/artifact-management.spec.ts:17	531ms
> ✓ Expect "toBeVisible" getByRole("heading", { name: 'Debug Lab' }) — artifacts/artifact-management.spec.ts:20	95ms
> ✓ POST "/api/cart/items" — artifacts/artifact-management.spec.ts:23	52ms
> ✓ GET "/api/cart" — artifacts/artifact-management.spec.ts:38	5ms
> ✓ Expect "toBe" — artifacts/artifact-management.spec.ts:47	0ms
> ✓ Expect "toBe" — artifacts/artifact-management.spec.ts:48	1ms
> ✓ Expect "toMatchObject" — artifacts/artifact-management.spec.ts:51	2ms
> ✓ After Hooks	683ms

Attachments

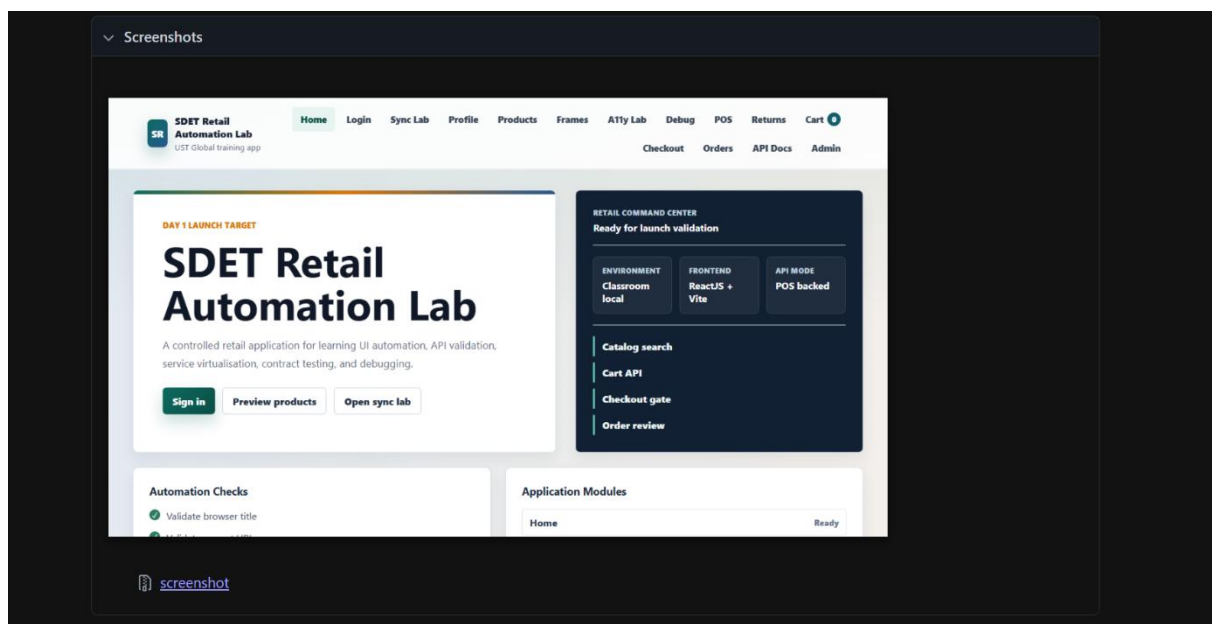
cartResponse.json

```
{
  "items": [
    {
      "id": 2,
      "userId": 1,
      "ownerKey": "user-1",
      "productId": 101,
      "quantity": 4,
      "size": "UK 9",
      "color": "Black",
      "fulfilment": "Home delivery",
      "product": {
        "id": 101,
        "name": "Running Shoes",
        "category": "Footwear",
        "price": 4499,
        "stock": 18,
        "brand": "SwiftRun",
        "sku": "FT-SHOE-101",
        "summary": "Lightweight daily trainers with breathable mesh and steady heel support.",
        "tags": [
          "Best seller",
          "Running",
          "COD eligible"
        ]
      }
    }
  ],
  "total": 17996
}
```

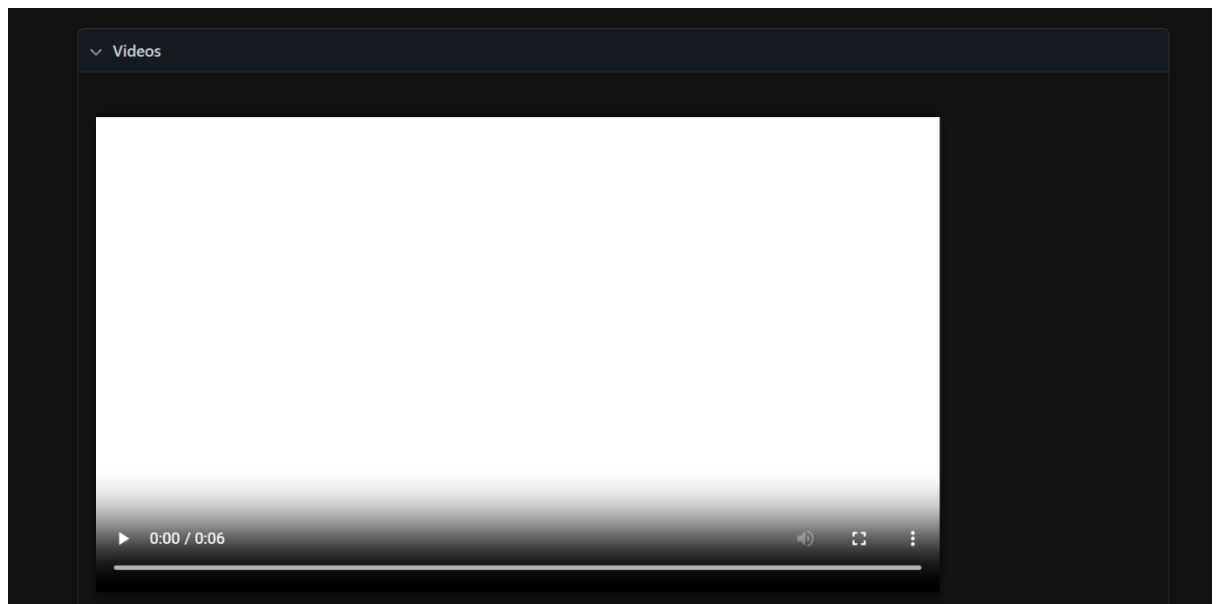
- HTML Report (Failure)



- Failure Screenshot



- Trace Back Screenshot



7. Offline & Resilient UI

Command to run in terminal:

```
npx playwright test network-offline.spec.ts
```

Test File: network-offline.spec.ts

Scenario 1

Verify offline banner.

Scenario 2

Queue sales while offline.

Scenario 3

Synchronize queued sales after reconnecting.

Scenario 4

Rollback optimistic UI after failed synchronization.

Scenario 5

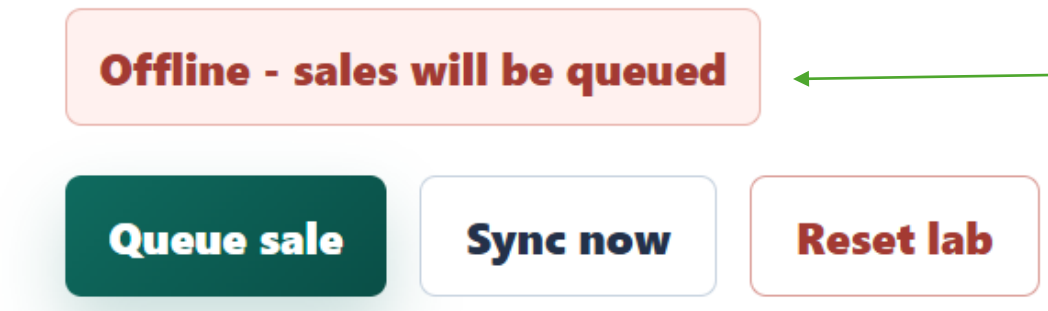
Ensure exactly-once synchronization using Idempotency.

Benefits

The application continues functioning correctly during network interruptions and safely recovers when connectivity is restored.

Evidence

- Offline banner screenshot



- Queue count screenshot



Pending outbox: **5** ←

- Sync completed screenshot



Pending outbox: **0**



- Rollback screenshot

Sync failed. Optimistic sale rolled back.



Pending outbox: **0**

Last synced sale: SALE-9010

- Exactly-once synchronization screenshot

Local outbox

Queued POS sales



CLIENT SALE ID	PRODUCT	STATUS	TOTAL
sale-1783159737845-a45ga8e4f85	Running Shoes	synced	Rs. 4,499
sale-1783159737057-no4212gvg6	Running Shoes	synced	Rs. 4,499
sale-1783159649279-q24xywyfq59	Running Shoes	synced	Rs. 4,499
sale-1783159648748-r89dj0do947	Running Shoes	synced	Rs. 4,499

8. Returns & Refund Testing

Test File: refund-return.spec.ts

The following business scenarios were validated.

Partial Refund

Command to run in terminal:

```
npx playwright test partial-refund.spec.ts
```

Verified:

- Refund amount
- Tax calculation
- Ledger update
- UI update

Evidence

- Before refund screenshot of UI

Returnable Lines				
Refund line ledger				
SKU	UNIT	PURCHASED	REFUNDED	RULE
TEE	₹333.00	3	0	RETURNABLE

- After refund screenshot of UI

Returnable Lines				
Refund line ledger				
SKU	UNIT	PURCHASED	REFUNDED	RULE
TEE	₹333.00	3	1	RETURNABLE

- Terminal Output

```
Running 1 test using 1 worker
[chromium] > tests\refunds\partial-refund.spec.ts:13:7 > Partial Refund > should successfully process a partial refund
[INFO] 2026-07-04T10:58:20.250Z - test started
[INFO] 2026-07-04T10:58:20.258Z - Seeding order
[INFO] 2026-07-04T10:58:20.391Z - Opening Returns page {"orderId":7020}
[INFO] 2026-07-04T10:58:21.378Z - Submitting partial refund
[INFO] 2026-07-04T10:58:21.441Z - Ledger verified {"status":"PARTIALLY_REFUNDED","refundCount":1}
[INFO] 2026-07-04T10:58:21.691Z - test finished {"status":"passed"}
1 passed (8.2s)
```

To open last HTML report run:

```
npx playwright show-report
```

```
C:\Users\299172\Documents\Saiteja_week05_gate5>
```

- HTML Report

Partial Refund

should successfully process a partial refund

refunds/partial-refund.spec.ts:13

3.3s

chromium

Run

Test Steps

Filter steps

> ✓ Before Hooks

1.6s

> ✓ POST "/api/refund-lab/orders" — ./src/api/refund.api.ts:14

101ms

> ✓ Expect "toBeTruthy" — ./src/api/refund.api.ts:32

1ms

> ✓ Navigate to "/returns?orderId=7020" — ./src/api/refund.api.ts:107

678ms

> ✓ Expect "toContainText" getByTestId('refund-line') — refunds/partial-refund.spec.ts:32

222ms

> ✓ Expect "toHaveText" getByTestId('refunded-TEE') — refunds/partial-refund.spec.ts:33

65ms

> ✓ POST "/api/refunds" — ./src/api/refund.api.ts:43

10ms

> ✓ Expect "toBeTruthy" — refunds/partial-refund.spec.ts:44

1ms

- Evidence

Attachments

order.json

```
{
  "id": 7020,
  "orderNumber": "RET-7020",
  "ownerKey": "user-1",
  "placedOn": "2026-06-29",
  "daysAgo": 5,
  "returnWindowDays": 30,
  "paymentMethod": "Original card",
  "lines": [
    {
      "sku": "TEE",
      "name": "Training Tee",
      "unitPaise": 33300,
      "qty": 3,
      "refundedQty": 0,
      "finalSale": false,
      "nonReturnable": false
    }
  ],
  "lineTotalPaise": 99900,
  "taxPaise": 4995,
  "totalPaise": 104895,
  "refundableBalancePaise": 104895,
  "refunds": [],
  "refundCount": 0,
  "lastRefund": null,
  "status": "PAID"
}
```

> refund.json

> ledger.json

Full Refund

Command to run in terminal:

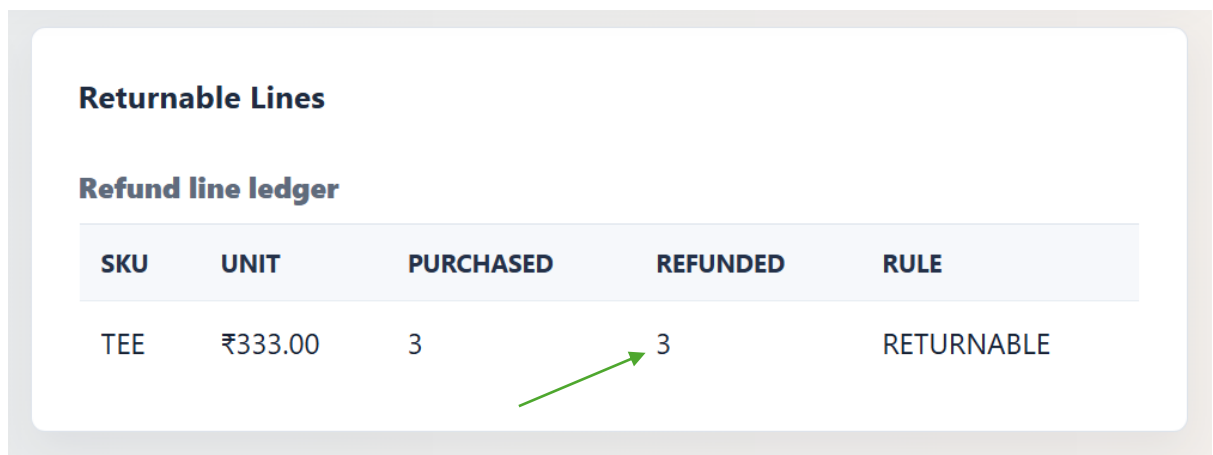
`npx playwright test full-refund.spec.ts`

Verified:

- Complete refund
- Refundable balance becomes zero
- Ledger updated correctly

Evidence

- Full refund screenshot of UI



Returnable Lines				
Refund line ledger				
SKU	UNIT	PURCHASED	REFUNDED	RULE
TEE	₹333.00	3	3	RETURNABLE

- Terminal Output

```
Running 1 test using 1 worker
[chromium] > tests\refunds\full-refund.spec.ts:13:7 > Full Refund > should successfully process a full refund
[INFO] 2026-07-04T11:03:36.847Z - test started
[INFO] 2026-07-04T11:03:36.856Z - Creating order
[INFO] 2026-07-04T11:03:37.020Z - Refunding entire order
[INFO] 2026-07-04T11:03:37.080Z - Full refund completed {"status":"REFUNDED"}
[INFO] 2026-07-04T11:03:38.296Z - test finished {"status":"passed"}
  1 passed (7.3s)
```

To open last HTML report run:

```
npx playwright show-report
```

```
C:\Users\299172\Documents\Saiteja_week05_gate5>
```

- HTML Report

Full Refund

should successfully process a full refund

refunds/full-refund.spec.ts:13 3.2s

chromium

✓ Run

Test Steps

Filter steps

> ✓ Before Hooks	1.6s
> ✓ POST "/api/refund-lab/orders" — ./src/api/refund.api.ts:14	120ms
> ✓ Expect "toBeTruthy" — ./src/api/refund.api.ts:32	4ms
> ✓ POST "/api/refunds" — ./src/api/refund.api.ts:43	14ms
> ✓ Expect "toBeTruthy" — refunds/full-refund.spec.ts:36	1ms
> ✓ Expect "toBe" — refunds/full-refund.spec.ts:42	1ms
> ✓ GET "/api/refund-lab/orders/7021" — ./src/api/refund.api.ts:91	8ms
> ✓ Expect "toBeTruthy" — ./src/api/refund.api.ts:98	0ms
> ✓ Expect "toBe" — refunds/full-refund.spec.ts:53	0ms

- Evidence

Attachments

order.json

```
{
  "id": 7021,
  "orderNumber": "RET-7021",
  "ownerKey": "user-1",
  "placedOn": "2026-06-29",
  "daysAgo": 5,
  "returnWindowDays": 30,
  "paymentMethod": "Original card",
  "lines": [
    {
      "sku": "TEE",
      "name": "Training Tee",
      "unitPaise": 33300,
      "qty": 3,
      "refundedQty": 0,
      "finalSale": false,
      "nonReturnable": false
    }
  ],
  "lineTotalPaise": 99900,
  "taxPaise": 4995,
  "totalPaise": 104895,
  "refundableBalancePaise": 104895,
  "refunds": [],
  "refundCount": 0,
  "lastRefund": null,
  "status": "PAID"
}
```

refund.json

ledger.json

Refund Eligibility

Command to run in terminal:

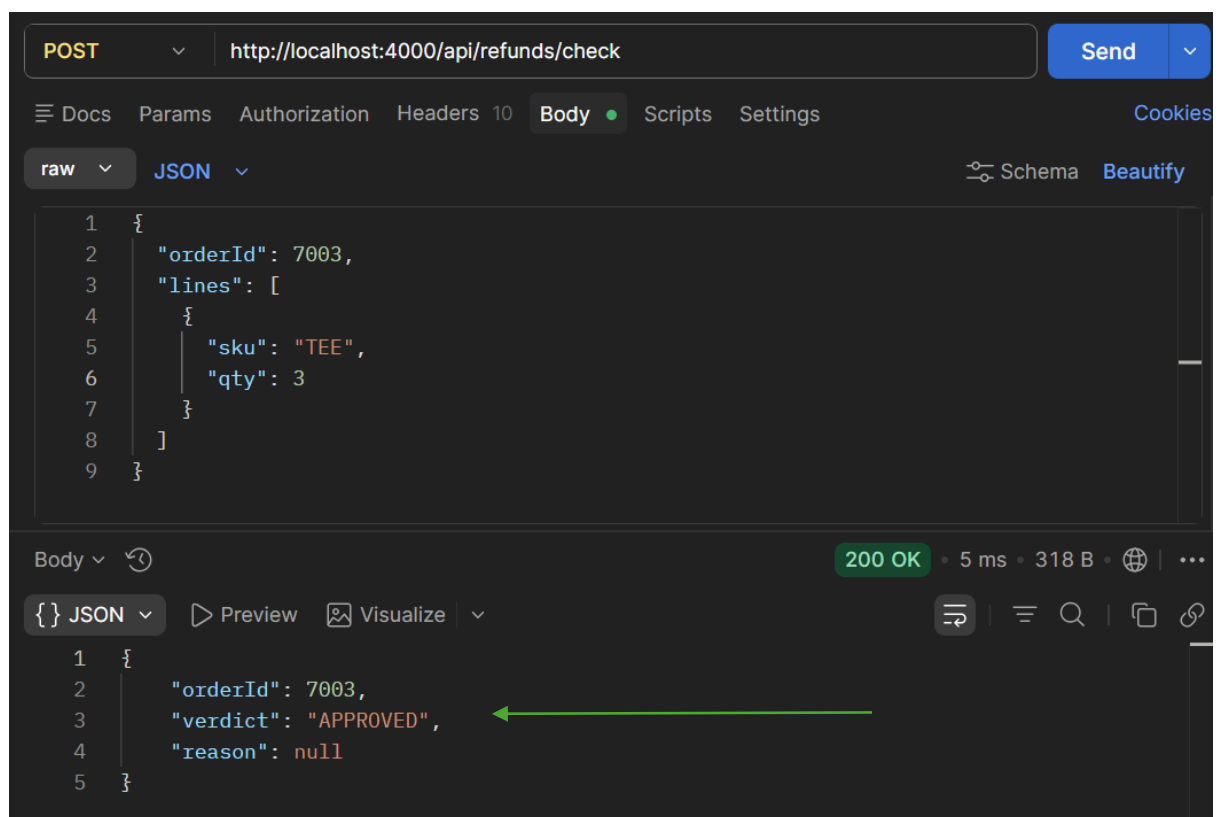
`npx playwright test refund-eligibility.spec.ts`

Verified:

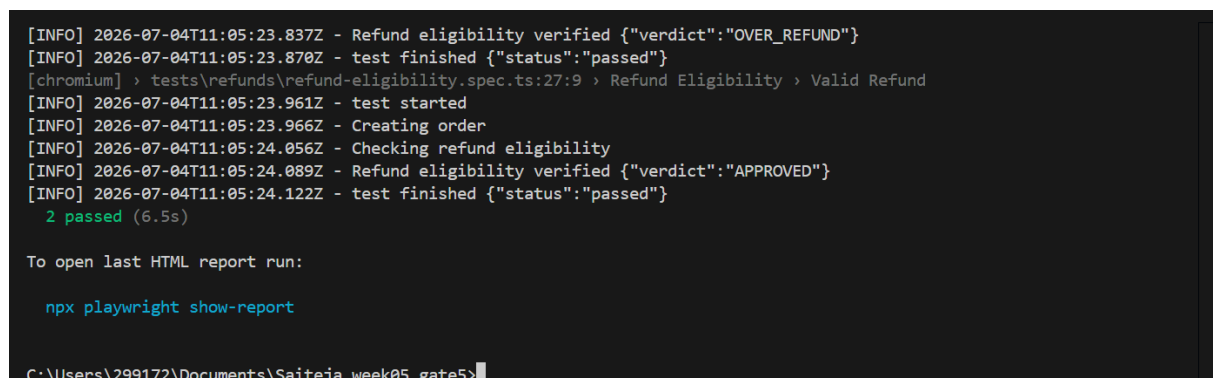
- Valid refund

Evidence

- Refund eligibility screenshot in postman



- Terminal Output



- HTML Report

Refund Eligibility

Valid Refund

refunds/refund-eligibility.spec.ts:27

2.4s

chromium

Run

Test Steps

Filter steps

> Before Hooks

2.2s

> POST "/api/refund-lab/orders" — ./src/api/refund.api.ts:14

65ms

> Expect "toBeTruthy" — ./src/api/refund.api.ts:32

1ms

> POST "/api/refunds/check" — ./src/api/refund.api.ts:66

9ms

> Expect "toBeTruthy" — ./src/api/refund.api.ts:82

0ms

> Expect "toBe" — refunds/refund-eligibility.spec.ts:52

1ms

> After Hooks

936ms

- Evidence

Attachments

order.json

```
{
  "id": 7023,
  "orderNumber": "RET-7023",
  "ownerKey": "user-1",
  "placedOn": "2026-06-29",
  "daysAgo": 5,
  "returnWindowDays": 30,
  "paymentMethod": "Original card",
  "lines": [
    {
      "sku": "TEE",
      "name": "Training Tee",
      "unitPaise": 33300,
      "qty": 3,
      "refundedQty": 0,
      "finalSale": false,
      "nonReturnable": false
    }
  ],
  "lineTotalPaise": 99900,
  "taxPaise": 4995,
  "totalPaise": 104895,
  "refundableBalancePaise": 104895,
  "refunds": [],
  "refundCount": 0,
  "lastRefund": null,
  "status": "PAID"
}
```

refundCheck.json

Rejected Refund

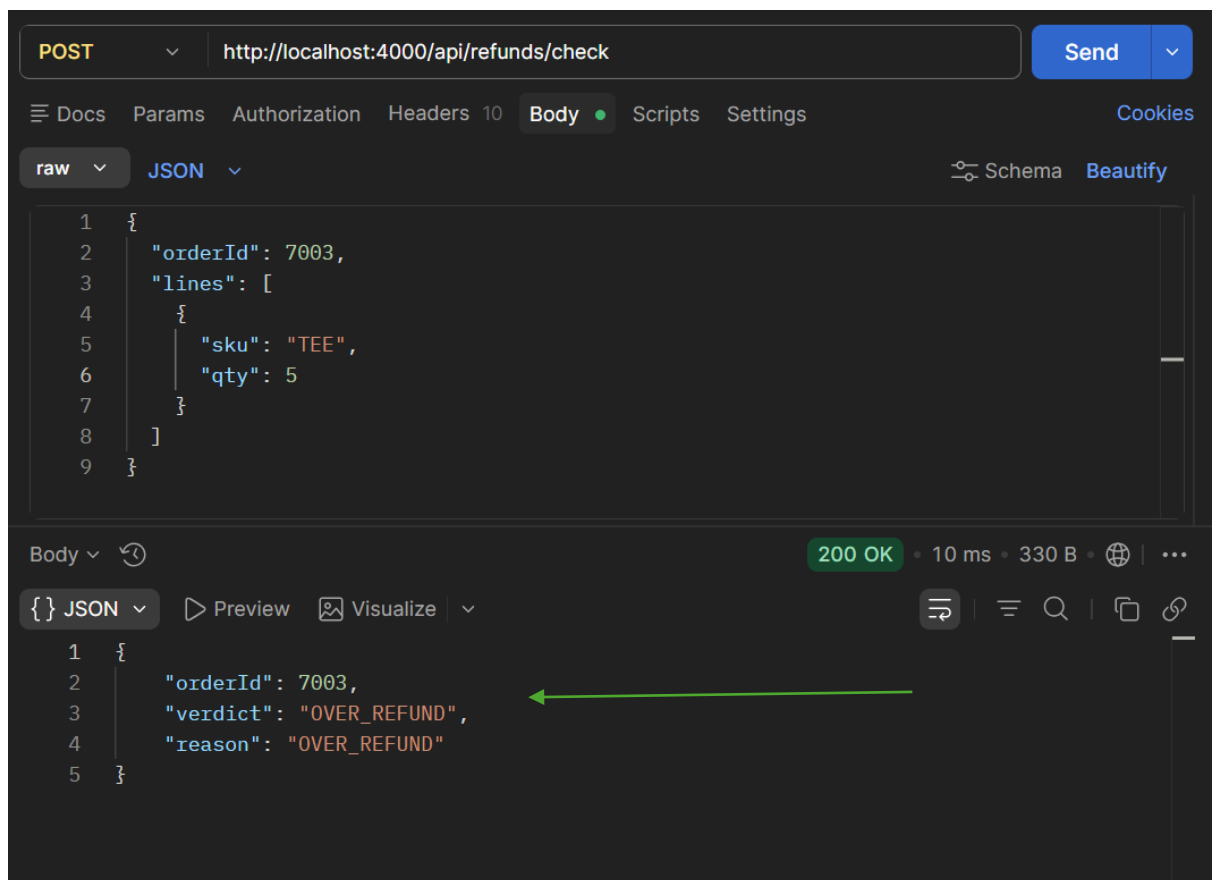
Command to run in terminal:

`npx playwright test rejected-refund.spec.ts`

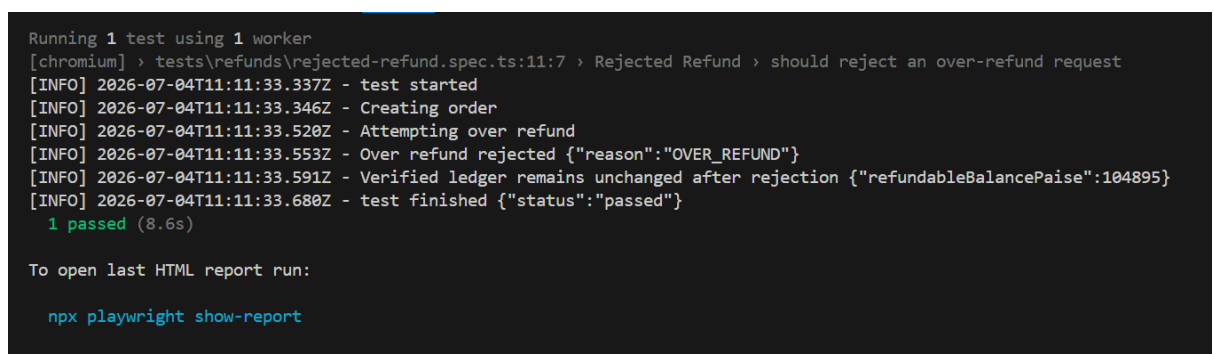
Verified that over-refund requests are rejected.

Evidence

- Rejected refund screenshot from postman



- Terminal Output



- HTML Report

The screenshot shows a Playwright HTML report for a test titled "should reject an over-refund request". The test is located at `refunds/rejected-refund.spec.ts:11` and took 1.9s to complete. It was run using Chromium. The report includes a "Test Steps" section with a search bar and a list of steps, each with a status icon, a description, and a duration.

Step	Duration
Before Hooks	1.8s
POST "/api/refund-lab/orders" — ./src/api/refund.api.ts:14	108ms
Expect "toBeTruthy" — ./src/api/refund.api.ts:32	2ms
GET "/api/refund-lab/orders/7024" — ./src/api/refund.api.ts:91	11ms
Expect "toBeTruthy" — ./src/api/refund.api.ts:98	0ms
POST "/api/refunds" — ./src/api/refund.api.ts:43	11ms
Expect "toBe" — refunds/rejected-refund.spec.ts:39	2ms
Expect "toBe" — refunds/rejected-refund.spec.ts:49	1ms
Expect "toBe" — refunds/rejected-refund.spec.ts:50	1ms

- Evidence

The screenshot shows the "Attachments" section of the Playwright HTML report. It contains a list of attachments, with "order.json" expanded to show its contents. The JSON data represents an order with various fields including id, orderNumber, ownerKey, placedOn, daysAgo, returnWindowDays, paymentMethod, lines, lineTotalPaise, taxPaise, totalPaise, refundableBalancePaise, refunds, refundCount, lastRefund, and status.

```
{
  "id": "7024",
  "orderNumber": "RET-7024",
  "ownerKey": "user-1",
  "placedOn": "2026-06-29",
  "daysAgo": 5,
  "returnWindowDays": 30,
  "paymentMethod": "Original card",
  "lines": [
    {
      "sku": "TEE",
      "name": "Training Tee",
      "unitPaise": 33300,
      "qty": 3,
      "refundedQty": 0,
      "finalSale": false,
      "nonReturnable": false
    }
  ],
  "lineTotalPaise": 99900,
  "taxPaise": 4995,
  "totalPaise": 104895,
  "refundableBalancePaise": 104895,
  "refunds": [],
  "refundCount": 0,
  "lastRefund": null,
  "status": "PAID"
}
```

Below the expanded "order.json" file, there are links to other attachments: `ledgerBefore.json`, `overRefundError.json`, and `ledgerAfter.json`.

Duplicate Refund

Command to run in terminal:

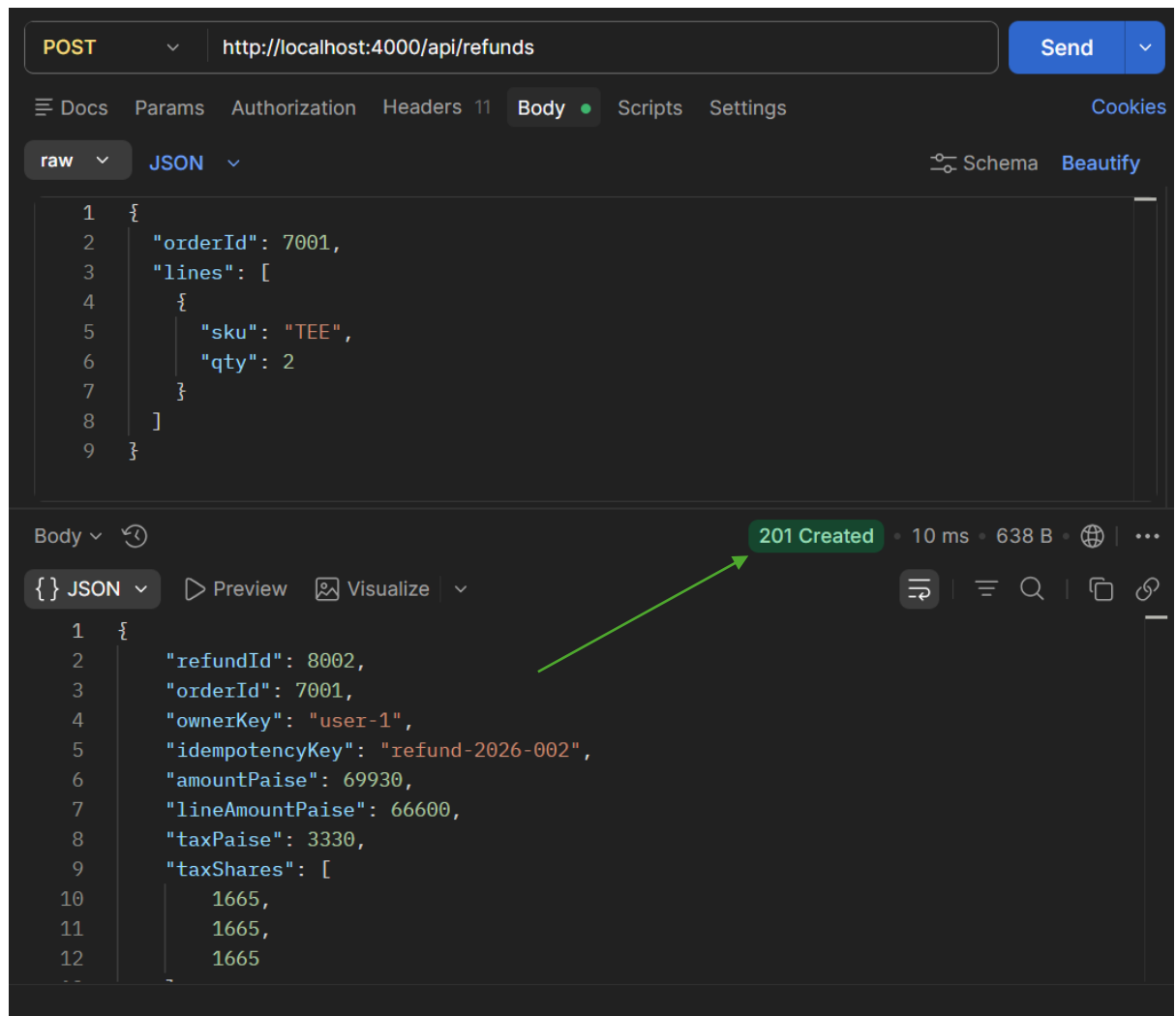
`npx playwright test duplicate-refund-idempotency.spec.ts`

The same refund request was submitted twice.

The application returned the existing refund instead of creating another one.

Evidence

- Duplicate refund
- Screenshot of First time request(Status Code = 201 Created) from Postman



- Terminal Output

```
Running 1 test using 1 worker
...efund-idempotency.spec.ts:12:7 › Duplicate Refund and Idempotency › should process only one refund for duplicate requests
[INFO] 2026-07-04T11:17:13.756Z - test started
[INFO] 2026-07-04T11:17:13.760Z - Creating order
[INFO] 2026-07-04T11:17:13.854Z - Submitting first refund request
[INFO] 2026-07-04T11:17:13.870Z - Submitting duplicate refund request
[INFO] 2026-07-04T11:17:13.901Z - Verified idempotent refund {"refundCount":1,"refundableBalancePaise":69930}
[INFO] 2026-07-04T11:17:13.986Z - test finished {"status":"passed"}
  1 passed (5.6s)

To open last HTML report run:

npx playwright show-report
```

- HTML Report

Duplicate Refund and Idempotency

should process only one refund for duplicate requests

refunds/duplicate-refund-idempotency.spec.ts:12

1.6s

chromium

✓ Run

Test Steps

Q Filter steps

> ✓ Before Hooks1.4s

> ✓ POST "/api/refund-lab/orders" — ./src/api/refund.api.ts:1477ms

> ✓ Expect "toBeTruthy" — ./src/api/refund.api.ts:321ms

> ✓ POST "/api/refunds" — ./src/api/refund.api.ts:438ms

> ✓ Expect "toBeTruthy" — refunds/duplicate-refund-idempotency.spec.ts:380ms

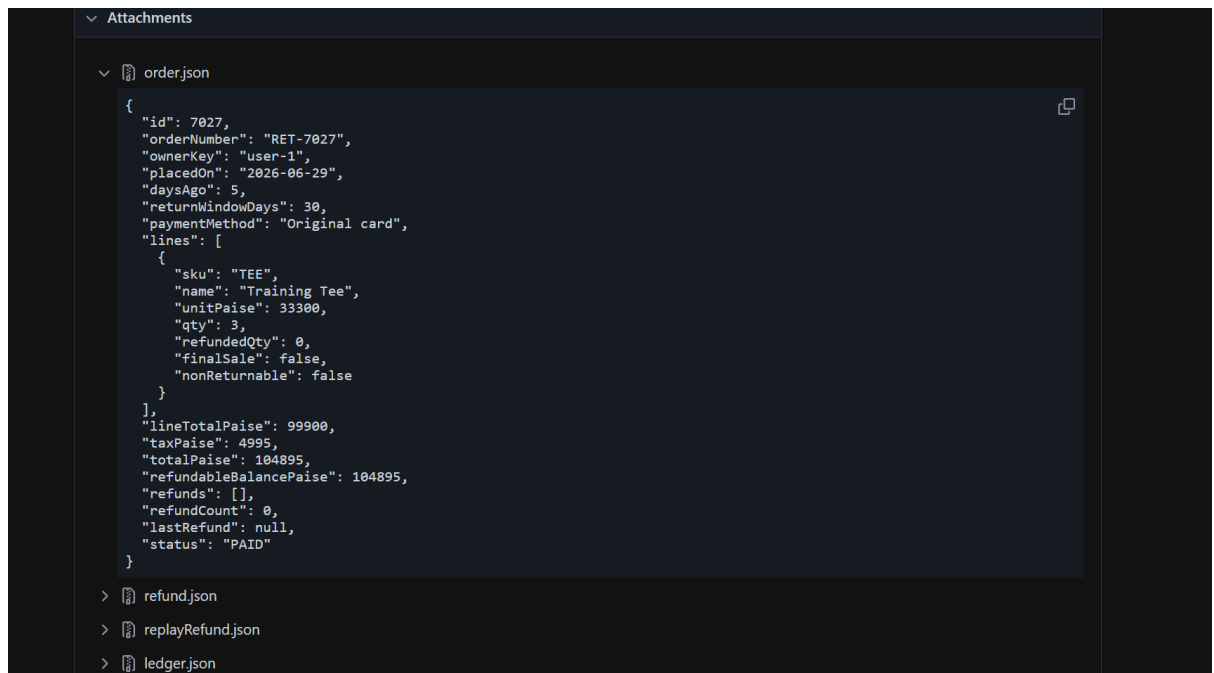
> ✓ POST "/api/refunds" — ./src/api/refund.api.ts:438ms

> ✓ Expect "toBe" — refunds/duplicate-refund-idempotency.spec.ts:531ms

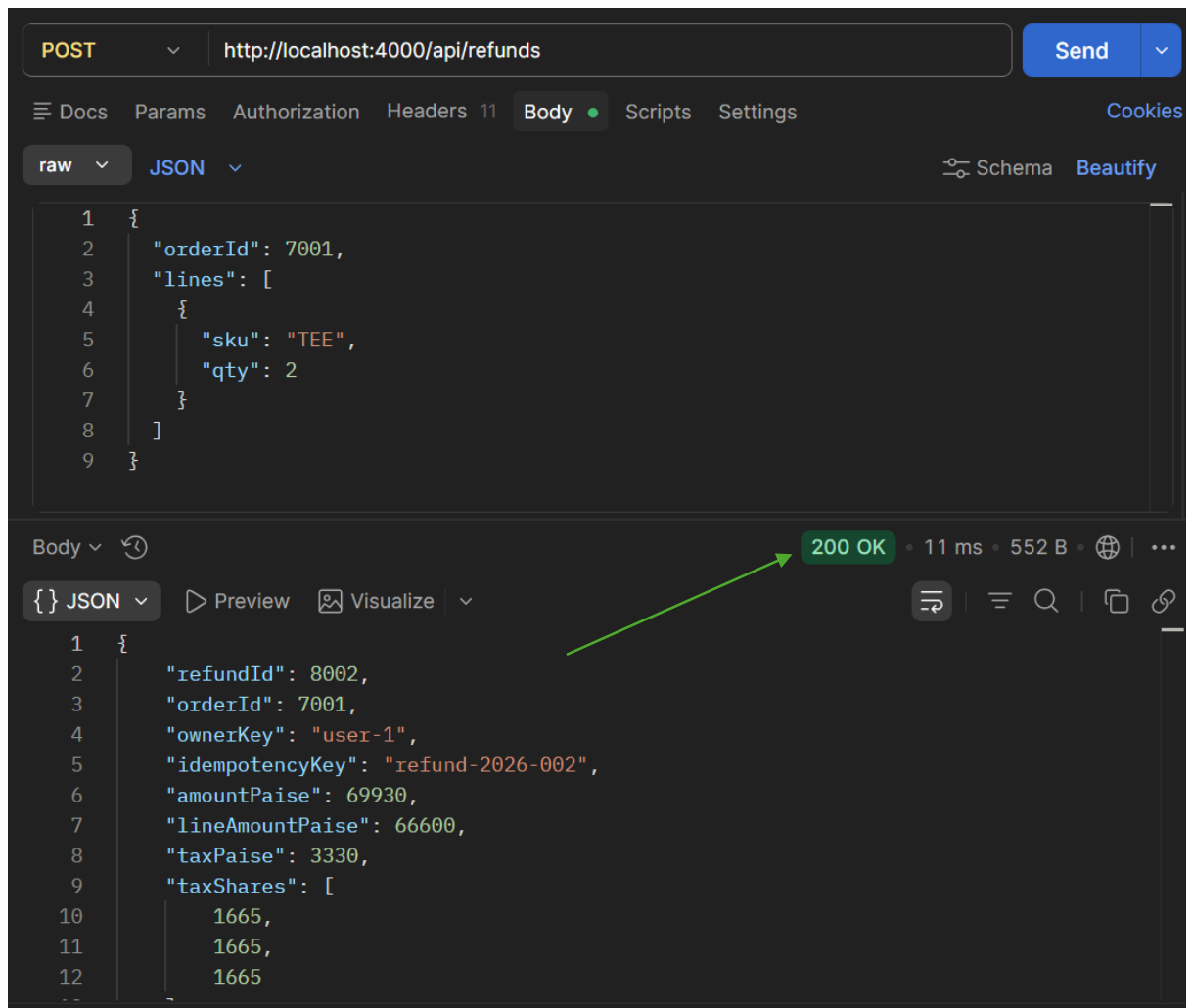
> ✓ Expect "toBe" — refunds/duplicate-refund-idempotency.spec.ts:590ms

> ✓ Expect "toBe" — refunds/duplicate-refund-idempotency.spec.ts:600ms

- Evidence



- Screenshot of Second time request(Status Code = 200 OK) from Postman



- Terminal Output

```
Running 1 test using 1 worker
...refund-idempotency.spec.ts:12:7 > Duplicate Refund and Idempotency > should process only one refund for duplicate requests
[INFO] 2026-07-04T11:17:13.756Z - test started
[INFO] 2026-07-04T11:17:13.760Z - Creating order
[INFO] 2026-07-04T11:17:13.854Z - Submitting first refund request
[INFO] 2026-07-04T11:17:13.870Z - Submitting duplicate refund request
[INFO] 2026-07-04T11:17:13.901Z - Verified idempotent refund {"refundCount":1,"refundableBalancePaise":69930}
[INFO] 2026-07-04T11:17:13.986Z - test finished {"status":"passed"}
1 passed (5.6s)

To open last HTML report run:

npx playwright show-report
```

- HTML Report

Duplicate Refund and Idempotency

should process only one refund for duplicate requests

refunds/duplicate-refund-idempotency.spec.ts:12

1.6s

chromium

✓ Run

Test Steps

Q Filter steps

> ✓ Before Hooks

1.4s

> ✓ POST "/api/refund-lab/orders" — ./src/api/refund.api.ts:14

77ms

> ✓ Expect "toBeTruthy" — ./src/api/refund.api.ts:32

1ms

> ✓ POST "/api/refunds" — ./src/api/refund.api.ts:43

8ms

> ✓ Expect "toBeTruthy" — refunds/duplicate-refund-idempotency.spec.ts:53

0ms

> ✓ POST "/api/refunds" — ./src/api/refund.api.ts:43

8ms

> ✓ Expect "toBe" — refunds/duplicate-refund-idempotency.spec.ts:53

1ms

> ✓ Expect "toBe" — refunds/duplicate-refund-idempotency.spec.ts:59

0ms

> ✓ Expect "toBe" — refunds/duplicate-refund-idempotency.spec.ts:60

0ms

- Evidence

Attachments

order.json

```
{
  "id": 7027,
  "orderNumber": "RET-7027",
  "ownerKey": "user-1",
  "placedOn": "2026-06-29",
  "daysAgo": 5,
  "returnWindowDays": 30,
  "paymentMethod": "Original card",
  "lines": [
    {
      "sku": "TEE",
      "name": "Training Tee",
      "unitPaise": 33300,
      "qty": 3,
      "refundedQty": 0,
      "finalSale": false,
      "nonReturnable": false
    }
  ],
  "lineTotalPaise": 99900,
  "taxPaise": 4995,
  "totalPaise": 104895,
  "refundableBalancePaise": 104895,
  "refunds": [],
  "refundCount": 0,
  "lastRefund": null,
  "status": "PAID"
}
```

> refund.json

> replayRefund.json

> ledger.json

Idempotency

Verified that duplicate requests with the same idempotency key are processed only once.

Evidence

- Idempotency screenshot

The screenshot shows a Postman interface for a POST request to `http://localhost:4000/api/refunds`. The Headers tab is active, displaying the following headers:

Key	Value	Description
Host	<calculated when request is sent>	
User-Agent	PostmanRuntime/7.54.0	
Accept	*/*	
Accept-Encoding	gzip, deflate, br	
Connection	keep-alive	
Authorization	Bearer demo-token-1-customer	
Content-Type	application/json	
Idempotency-Key	refund-2026-002	

The Body tab shows a JSON response:

```
{  "refundId": 8002,  "orderId": 7001,  "ownerKey": "user-1",  "idempotencyKey": "refund-2026-002",  "amountPaise": 69930,  "lineAmountPaise": 66600,  "taxPaise": 3330,  "taxShares": [    1665,    1665,    1665  ]}
```

The status bar indicates a **200 OK** response with a time of 11 ms and a body size of 552 B. A green arrow points to the status bar.

- Terminal Output

```
Running 1 test using 1 worker
...efund-idempotency.spec.ts:12:7 > Duplicate Refund and Idempotency > should process only one refund for duplicate requests
[INFO] 2026-07-04T11:17:13.756Z - test started
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[INFO] 2026-07-04T11:17:13.870Z - Submitting duplicate refund request
[INFO] 2026-07-04T11:17:13.901Z - Verified idempotent refund {"refundCount":1,"refundableBalancePaise":69930}
[INFO] 2026-07-04T11:17:13.986Z - test finished {"status":"passed"}
1 passed (5.6s)

To open last HTML report run:

npx playwright show-report
```

- HTML Report

Duplicate Refund and Idempotency

should process only one refund for duplicate requests

refunds/duplicate-refund-idempotency.spec.ts:12

1.6s

chromium

✓ Run

Test Steps

Q Filter steps

> ✓ Before Hooks1.4s

> ✓ POST "/api/refund-lab/orders" — ./src/api/refund.api.ts:1477ms

> ✓ Expect "toBeTruthy" — ./src/api/refund.api.ts:321ms

> ✓ POST "/api/refunds" — ./src/api/refund.api.ts:438ms

> ✓ Expect "toBeTruthy" — refunds/duplicate-refund-idempotency.spec.ts:380ms

> ✓ POST "/api/refunds" — ./src/api/refund.api.ts:438ms

> ✓ Expect "toBe" — refunds/duplicate-refund-idempotency.spec.ts:531ms

> ✓ Expect "toBe" — refunds/duplicate-refund-idempotency.spec.ts:590ms

> ✓ Expect "toBe" — refunds/duplicate-refund-idempotency.spec.ts:600ms

- Evidence

Attachments

order.json

```
{
  "id": 7027,
  "orderNumber": "RET-7027",
  "ownerKey": "user-1",
  "placedOn": "2026-06-29",
  "daysAgo": 5,
  "returnWindowDays": 30,
  "paymentMethod": "Original card",
  "lines": [
    {
      "sku": "TEE",
      "name": "Training Tee",
      "unitPaise": 33300,
      "qty": 3,
      "refundedQty": 0,
      "finalSale": false,
      "nonReturnable": false
    }
  ],
  "lineTotalPaise": 99900,
  "taxPaise": 4995,
  "totalPaise": 104895,
  "refundableBalancePaise": 104895,
  "refunds": [],
  "refundCount": 0,
  "lastRefund": null,
  "status": "PAID"
}
```

> refund.json

> replayRefund.json

> ledger.json

Money Precision

Verified all monetary calculations using **paise** instead of floating-point values.

Validated:

- Refund Amount
- Tax
- Ledger Balance

Evidence

- Money precision screenshot

```
"amountPaise": 69930,
"lineAmountPaise": 66600,
"taxPaise": 3330,
"taxShares": [
  1665,
  1665,
  1665
],
```

9. Failure Analysis

Every issue in this assessment follows the same debugging approach.

Symptom

Identify the observed failure.

Diagnosis

Analyze logs, assertions and application behavior to determine the root cause.

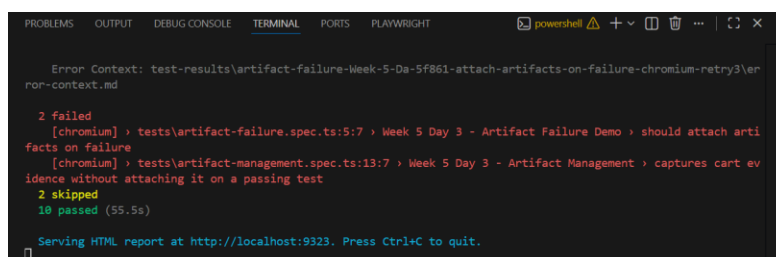
Fix

Apply the appropriate solution instead of temporary workarounds.

Evidence

Validate the fix using:

- Screenshot



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS PLAYWRIGHT
Error Context: test-results\artifact-failure-Week-5-Da-5f861-attach-artifacts-on-failure-chromium-retry3\error-context.md

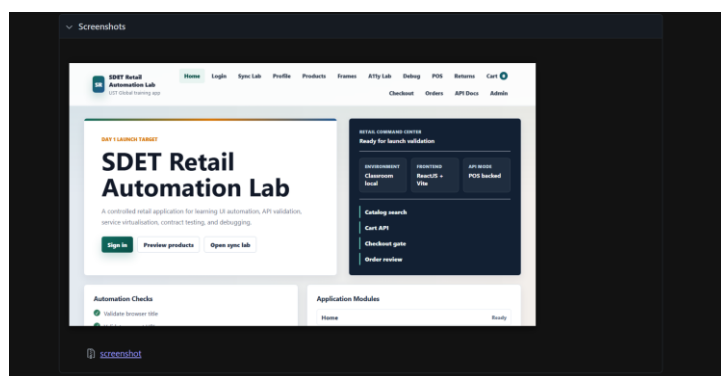
2 failed
[chromium] › tests\artifact-failure.spec.ts:5:7 › Week 5 Day 3 - Artifact Failure Demo › should attach artifacts on failure
[chromium] › tests\artifact-management.spec.ts:13:7 › Week 5 Day 3 - Artifact Management › captures cart evidence without attaching it on a passing test
2 skipped
10 passed (55.5s)

Serving HTML report at http://localhost:9323. Press Ctrl+C to quit.
```

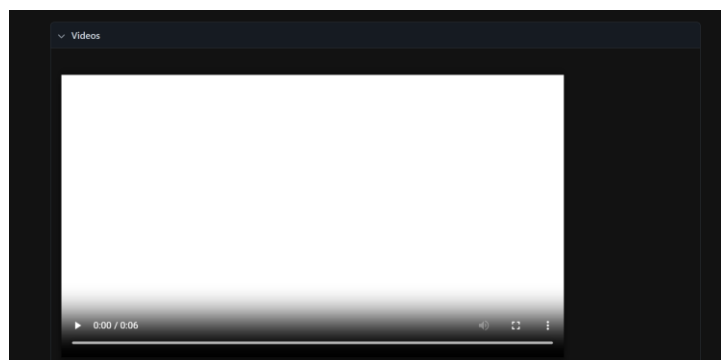
- HTML Report



- Failure Screenshot



- Trace Back Video



10. Debugging Maturity

This assessment demonstrates debugging maturity through:

- Identifying flaky tests
- Diagnosing root causes
- Structured diagnostic logging
- Artifact collection

- Offline resilience testing
 - Business rule validation
 - Refund validation
 - Idempotency testing
 - Financial validation using paise
-

11. Challenges Faced

- Diagnosing flaky tests
 - Handling offline scenarios
 - Validating idempotent requests
 - Preventing duplicate refunds
 - Testing financial calculations
 - Capturing useful debugging evidence
-

12. Conclusion

This assessment successfully demonstrates stable Playwright automation by focusing on diagnosing issues, implementing reliable solutions, collecting meaningful evidence, and clearly explaining the results. The project validates critical business scenarios such as offline operations, refund processing, artifact management, and structured diagnostic logging while improving overall debugging maturity.

13. Evidence Summary

Assessment Area	Evidence to Include
Flakiness Analysis	Before failure screenshot, After fix screenshot, HTML Report
Diagnostic Logging	Correlation ID logs, Structured logs, Redacted logs, Trace flow screenshot

Assessment Area	Evidence to Include
Artifact Management	HTML Report (Pass), HTML Report (Fail), Failure Screenshot, Trace Back Screenshot
Offline & Resilient UI	Offline banner, Queue count, Rollback, Sync success
Returns & Refund Testing	Partial refund, Full refund, Rejected refund, Duplicate refund, Idempotency, Money precision